ECA1308 – MICROWAVE ENGINEERING

CO1 Application-Based Assessment

Analysis, Design and Verification of a Microwave Transmission Line and Network for a Selected Application

Selected application: X-band radar front-end feed using a rectangular copper waveguide (WR-90)

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Abstract

This report develops an analytical design for an X-band radar front-end feed using a rectangular copper waveguide. The selected nominal operating frequency is 10 GHz, and the guide dimensions are a = 22.86 mm and b = 10.16 mm, corresponding to the standard WR-90 size. The design objective is to carry microwave energy with low conductor loss, controlled impedance, and operation in the dominant TE10 mode. The report defines application requirements and compares candidate transmission structures. It calculates cutoff frequencies, guide wavelength, phase and group velocities, wave impedance, conductor surface resistance and estimated attenuation. A two-port equivalent is used to discuss reflection, VSWR, return loss, insertion loss and matching. Validation considers mode margin, manufacturing tolerance, EMI/EMC and limitations of the analytical model. Further verification using a microwave network analyser or full-wave solver is required before hardware adoption.

Keywords: X-band radar, rectangular waveguide, TE10 mode, cutoff frequency, attenuation, S-parameters, impedance matching.

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1. Stage 1 – Formulate the Application and Operating Requirement (10 marks)

1.1 Application and category

The chosen application is the feed between an X-band radar front-end and its antenna or waveguide subsystem. It is classified as a civil/military sensing application; the calculations here are generic and do not describe a particular operational radar. A waveguide is suitable where a microwave signal must be transported at high frequency with low conductor loss and good shielding. A rectangular guide also provides a clear dominant-mode design target.

1.2 Brief microwave technology context

Microwave engineering grew from electromagnetic theory and experimental demonstrations of electromagnetic waves, followed by practical radio systems. The cavity magnetron and radar accelerated high-power microwave development during the twentieth century. Later advances in satellite communications, semiconductor microwave devices, monolithic microwave integrated circuits (MMICs), and modern 5G/6G systems expanded the need for compact, low-loss and well-matched transmission structures. Waveguides remain useful in high-frequency and higher-power assemblies, while planar lines are convenient for integrated circuits.

1.3 Frequency-band comparison

Band	Approx. range used here	Wavelength at representative frequency	Advantages	Limitations
S band	2–4 GHz	At 3 GHz: 100 mm	Lower free-space/rain sensitivity than higher bands; larger wavelength	Larger components for a given guided design
X band	8–12 GHz	At 10 GHz: 30.0 mm	Compact antennas/feeds; established radar use; practical waveguide components	Greater atmospheric/rain effects than lower bands; precision alignment required
Ka band	26.5–40 GHz	At 30 GHz: 10.0 mm	Small components and potentially wide bandwidth	Higher rain/atmospheric attenuation; tighter fabrication tolerances

The X band is selected because it balances component size and practical radar feed implementation. Actual frequency allocation, permitted bandwidth and atmospheric attenuation depend on the intended jurisdiction and system; this report uses 10 GHz as a design point rather than claiming regulatory approval.

1.4 Quantified requirement specification

Parameter	Design target / assumption	Acceptance criterion
Application	X-band radar feed	Transmit RF energy between front-end and antenna subsystem
Operating frequency	10.0 GHz nominal	Remain inside calculated single-mode band
Waveguide	Rectangular copper WR-90; a = 22.86 mm, b = 10.16 mm	TE10 dominant mode
Reference impedance	TE modal impedance at 10 GHz, calculated below	Use consistently for wave variables and matching
Insertion loss	Target ≤ 0.5 dB over 0.30 m (design target)	Analytical conductor loss plus transition/mismatch losses must meet target
Return loss	Target ≥ 20 dB after matching	$\|S11\| \leq 0.1$ over the stated band
VSWR	Target ≤ 1.22 after matching	Calculated from reflection coefficient
Isolation / shielding	No universal numerical target assumed; measure in the assembly	Set application-specific EMI/EMC target before hardware test
Power	Low-power bench verification first; final power set by component ratings	No operation beyond flange, load and waveguide ratings
Size and cost	Standard WR-90 geometry; cost depends on supplier and flange/finish	Use actual supplier quote and mechanical drawing

1.5 EMI/EMC, safety and environment

A metal waveguide provides a conductive enclosure that generally limits radiated leakage compared with an unshielded planar line, but leakage can occur at flange joints, damaged surfaces and poorly fitted transitions. Use sound conductive flanges, correct fasteners, clean mating faces and appropriate grounding/bonding. Start with low-power testing and never look into an energized microwave port. High-power systems require a formal exposure assessment, interlocks, rated loads and approved procedures. Environmental considerations include corrosion, temperature variation, moisture ingress and material recyclability.

Stage 1 deliverable: a quantified specification and explicit acceptance criteria. Values labelled as targets are design assumptions and should be revised to match the actual project.

2. Stage 2 – Evaluate and Select the Transmission Structure (15 marks)

2.1 Mode and field comparison

Structure	Wave type / mode	Cutoff	Strengths	Limitations
Coaxial line	TEM dominant; higher modes possible above their onset	TEM has no cutoff; higher modes have limits	Flexible, shielded, connectorized	Loss can rise at high frequency; connector and conductor quality matter
Rectangular waveguide	TE/TM modes; TE10 dominant for a > b	Has cutoff for each mode; TE10 cutoff calculated in Stage 3	Low loss and good shielding at microwave frequencies	Bulkier, mechanically precise, flange transitions needed
Stripline	TEM (ideal homogeneous dielectric)	No TEM cutoff in ideal model	Shielded planar integration	Dielectric and conductor loss; multilayer fabrication
Microstrip	Quasi-TEM because fields occupy substrate and air	No simple hollow-guide cutoff for fundamental quasi-TEM mode	Compact, inexpensive and easy PCB integration	Radiation, dispersion and dielectric/substrate dependence

A hollow single-conductor waveguide cannot support TEM because a TEM field requires both electric and magnetic fields transverse to propagation and a suitable two-conductor potential structure. Microstrip is not pure TEM because some electric field exists in air and some in the dielectric, so the field distribution and effective permittivity are frequency-dependent.

2.2 Weighted decision matrix

Scores are engineering judgements from 1 (poor) to 5 (excellent) for this radar-feed example. Weights sum to 100%; they are not universal ratings.

Criterion	Weight	Coax	Waveguide	Stripline	Microstrip
Low loss at 10 GHz	25%	3	5	3	3
Power handling	15%	3	5	3	2
Shielding / EMI	15%	4	5	4	2
Size / weight	10%	4	2	4	5
Cost	10%	4	2	3	5
Integration / transitions	15%	4	3	5	5
Bandwidth / mode control	10%	3	5	4	3
Weighted score	100%	3.50/5	4.10/5	3.65/5	3.40/5

Under these weights, scores are coaxial 3.50/5, waveguide 4.10/5, stripline 3.65/5 and microstrip 3.40/5. Waveguide ranks first because the selected application prioritizes low loss, shielding and power handling. For a compact low-cost PCB receiver, increasing the weights on integration, size and cost could make microstrip rank first. At substantially lower frequency, coaxial cable may become more attractive; at higher frequency or higher power, waveguide can remain advantageous, subject to mode and mechanical constraints.

2.3 Decision and frequency change

Design Decision 1: For this X-band radar feed, select rectangular waveguide over microstrip because the assumed priorities are low loss, conductive enclosure and power handling. Microstrip is the strongest alternative where direct PCB integration, low cost and small footprint dominate. If the frequency is halved to 5 GHz, WR-90 is below TE10 cutoff and will not carry the intended propagating mode. If frequency is doubled to 20 GHz, higher-order modes can propagate and the standard WR-90 guide is no longer single-mode. The selected structure therefore depends on both frequency and dimensions.

3. Stage 3 – Develop the Mode and Propagation Model (20 marks)

3.1 Transmission-line equations

For a uniform line with per-unit-length resistance R, inductance L′, conductance G and capacitance C, the telegrapher's equations are $\partial V/\partial z = -(R + j\omega L')I$ and $\partial I/\partial z = -(G + j\omega C)V$. The propagation constant is $\gamma = \alpha + j\beta = \sqrt{[(R + j\omega L')(G + j\omega C)]}$, and characteristic impedance is $Z0 = \sqrt{[(R + j\omega L')/(G + j\omega C)]}$. For a low-loss line, $\alpha \approx (R/2)\sqrt{(C/L')} + (G/2)\sqrt{(L'/C)}$, while $\beta \approx \omega\sqrt{(L'C)}$. These equations provide the equivalent circuit view; a waveguide additionally requires field-mode analysis.

3.2 Rectangular waveguide cutoff

For a rectangular guide of width a and height b filled with air, the TE/TM cutoff frequency is $fc(mn) = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$. The dominant TE10 mode has $fc10 = c/(2a)$. TE20 has $fc20 = c/a$, and TE01 has $fc01 = c/(2b)$.

Parameter	Formula	Calculated value
Width a	Standard WR-90 dimension	22.86 mm
Height b	Standard WR-90 dimension	10.16 mm
TE10 cutoff	c/(2a)	6.557 GHz
TE20 cutoff	c/a	13.114 GHz
TE01 cutoff	c/(2b)	14.754 GHz
Free-space wavelength	c/f	29.979 mm
Phase constant β	$\sqrt{(k0^2 - (\pi/a)^2)}$	158.24 rad/m
Guide wavelength λg	2π/β	39.71 mm
Phase velocity vp	ω/β	3.971×10^8 m/s
Group velocity vg	c^2/vp	2.263×10^8 m/s
TE wave impedance	$\eta0 / \sqrt{[1 - (fc10/f)^2]}$	499.0 Ω
Group delay over 0.30 m	L/vg	1.325 ns

3.3 Worked calculation at 10 GHz

Using a = 0.02286 m: fc10 = c/(2a) = 299,792,458/(2 × 0.02286) = 6.557 GHz. Since 10 GHz > 6.557 GHz, TE10 propagates. The free-space wavenumber is $k0 = 2\pi f/c = 209.58$ rad/m. Then $\beta = \sqrt{[k0^2 - (\pi/a)^2]} = 158.24$ rad/m, so $\lambda g = 2\pi/\beta = 39.71$ mm. Phase velocity vp = 3.971e+08 m/s and group velocity vg = 2.263e+08 m/s.

3.4 Single-mode range and margin

The theoretical single-mode interval is 6.557 GHz < f < 13.114 GHz. Applying an illustrative 10% margin above TE10 cutoff and 10% below the first higher-mode cutoff gives a recommended interval of 7.213 to 11.803 GHz. The 10 GHz design point lies within this interval.

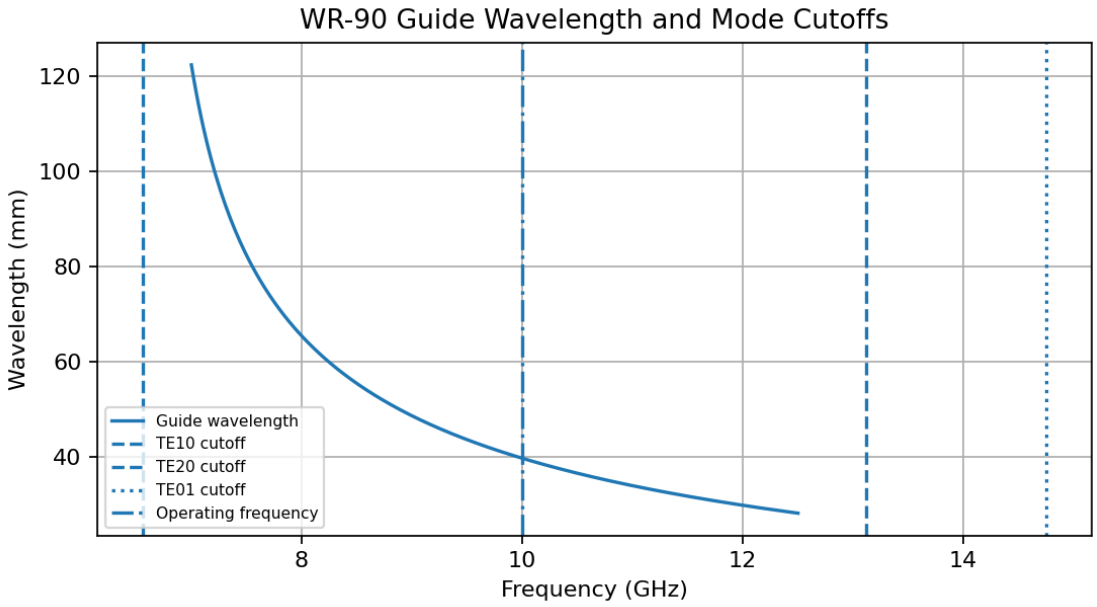


Figure 1. Analytical guide wavelength and modal cutoff markers for WR-90.

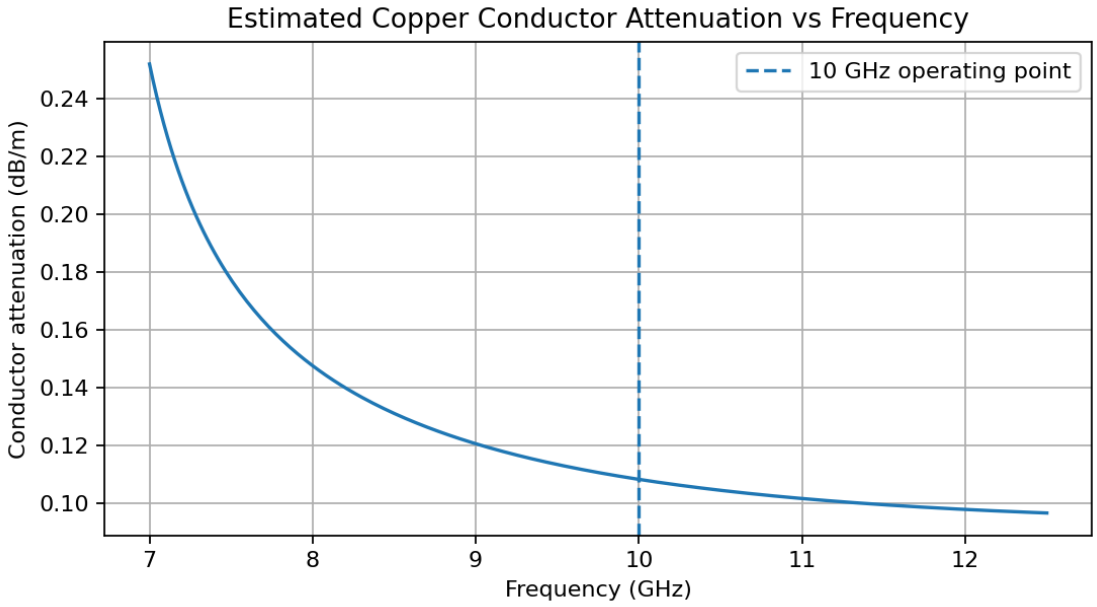


Figure 2. Analytical copper conductor attenuation trend; actual values depend on wall finish, conductivity, geometry and transitions.

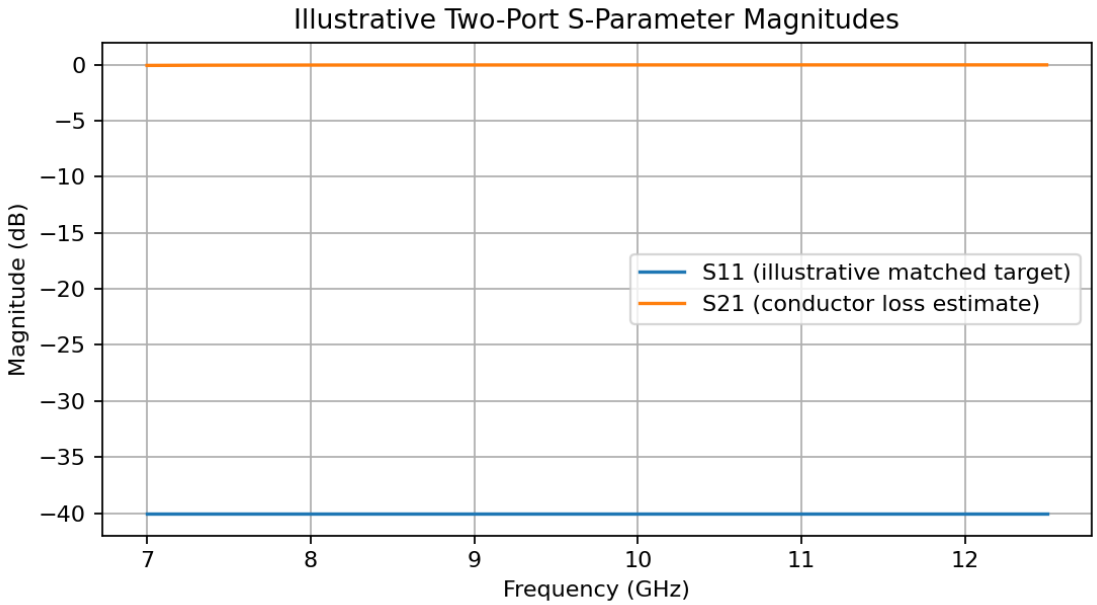


Figure 3. Illustrative analytical S-parameter magnitude targets; the S11 trace is a target/plotting floor, not measured reflection.

3.5 Higher-order modes and design decision

The first higher-mode cutoff is min(fc20, fc01) = 13.114 GHz. Feed transitions, flange misalignment, sharp discontinuities or asymmetric excitation may excite unwanted modes. Maintain the operating band inside the single-mode region, use a controlled launcher, avoid abrupt geometric changes, and verify flange alignment. Design Decision 2: the dominant mode is TE10; its cutoff is approximately 6.557 GHz; the recommended margin-based range is approximately 7.21–11.80 GHz; the 10 GHz dimensions are acceptable for the stated target.

4. Stage 4 – Evaluate Losses and Design a Loss-Reduction Strategy (15 marks)

4.1 Conductor loss calculation

Quantity	Formula	Result
Copper conductivity	Assumed material value	5.8×10^7 S/m
Skin depth δ	$\sqrt{[2/(2\pi f \mu 0\ \sigma)]}$	0.661 μm
Surface resistance Rs	$\sqrt{(\pi\ f\ \mu 0/\sigma)}$	0.0261 Ω/square
Conductor attenuation	Rectangular TE10 wall-loss expression	0.108 dB/m
0.30 m conductor loss	αc × L	0.033 dB

Worked estimate: $\delta = \sqrt{[2/(2\pi \times 10^{10} \times \mu 0 \times 5.8 \times 10^7)]} = 6.609\text{e-}07$ m = 0.661 μm. Surface resistance $Rs = \sqrt{(\pi \times 10^{10} \times \mu 0/(5.8 \times 10^7))} = 0.0261$ Ω/square. Using the standard TE10 rectangular-guide conductor attenuation expression gives an analytical estimate of 0.108 dB/m, or 0.033 dB over 0.30 m. This assumes smooth walls, uniform copper conductivity and an ideal straight guide.

4.2 Loss budget

Loss component	Initial analytical design	Improved design proposal	Data status / trade-off
Conductor	Calculated straight-guide loss: 0.033 dB over 0.30 m	Smoother inner walls or qualified low-roughness conductive finish	Improvement requires verified model/data
Dielectric	0 dB for ideal air-filled straight guide model	No dielectric added; preserve air-filled guide	Idealized straight section
Radiation / leakage	Not quantified analytically	Improve flange fit, bonding and transition design	Measure leakage / insertion loss
Connector / transition	Not quantified analytically	Use matched launcher and aligned flanges	Obtain datasheet or simulated S-parameters
Mismatch	Illustrative unmatched load calculated below	Use matching section / tune transition	Requires network analysis across band
Total	Only straight-section conductor component quantified	Final total requires all measured/simulated terms	Partial sum is not full assembly loss

4.3 Parametric trends and reduction strategy

Conductor attenuation depends on frequency through surface resistance and propagation constants. Increasing conductivity reduces surface resistance; rougher walls increase effective conductor loss, especially when roughness is comparable to skin depth. A dielectric-filled or planar structure adds dielectric loss proportional to loss tangent and field energy in the dielectric. For this air-filled waveguide, preferred measures are smoother high-conductivity internal surfaces, clean aligned flanges, and a well-designed transition. Any dB improvement must be quantified from a specified surface-finish model, supplier data or validated EM simulation.

Design Decision 3: the quantified straight-section loss is approximately 0.033 dB over 0.30 m. The proposed reduction is to improve wall finish and transition/flange matching. Exact improvement remains to be established by a verified model or measurement. The Stage 1 $\leq$0.5 dB target cannot be confirmed for the complete assembly until transition and mismatch losses are included.

5. Stage 5 – Create and Evaluate the Network Model (20 marks)

5.1 Equivalent voltage/current and modal impedance

For TEM lines, voltage and current can be defined by transverse field integrals and conductor currents. For non-TEM waveguides, unique scalar voltage/current are not inherently defined in the same way; equivalent circuit variables can be defined for a selected mode and normalization. For TE modes, modal wave impedance is $Z_{TE} = \omega\mu/\beta = \eta_0/\sqrt{[1 - (f_c/f)^2]}$. At 10 GHz, Z_{TE} is given in Stage 3.

5.2 Wave variables and S-parameters

For real reference impedance Z_0, normalized incident and reflected waves may be written $a = (V + Z_0 I)/(2\sqrt{Z_0})$ and $b = (V - Z_0 I)/(2\sqrt{Z_0})$, under a consistent port-current convention. The scattering relation is $[b_1\ b_2]^T = [S_{11}\ S_{12};\ S_{21}\ S_{22}][a_1\ a_2]^T$. S_{11} and S_{22} describe input/output reflection; S_{21} and S_{12} describe forward/reverse transmission. For a reciprocal passive structure, $S_{21} = S_{12}$ under consistent port definitions. A symmetric structure also has $S_{11} = S_{22}$. A lossless network satisfies $S^\dagger S = I$.

5.3 Illustrative load mismatch and matching

For an illustrative equivalent load $Z_L = 1.5\ Z_{TE} = 748.5\ \Omega$ and reference impedance $Z_0 = Z_{TE} = 499.0\ \Omega$, $\Gamma = (Z_L - Z_0)/(Z_L + Z_0) = 0.2000$. VSWR $= (1 + |\Gamma|)/(1 - |\Gamma|) = 1.500$; return loss $= -20 \log_{10}|\Gamma| = 13.98$ dB. This is a teaching example, not a measured radar load.

For a real load, a quarter-wave transformer uses $Z_t = \sqrt{(Z_0 Z_L)} = \sqrt{(499.0 \times 748.5)} = 611.1\ \Omega$ and an electrical length of $\lambda/4$ at the design frequency. In a physical waveguide, implement the equivalent target through a geometry-specific step, iris or matched transition and verify it; do not assume a scalar-impedance section directly maps to a manufacturable guide.

5.4 ABCD cascade and line section

For a uniform section with characteristic impedance Z_c, propagation constant γ and length l, its ABCD matrix is $[\cosh(\gamma l),\ Z_c \sinh(\gamma l);\ (1/Z_c)\sinh(\gamma l),\ \cosh(\gamma l)]$. Two cascaded sections combine by $M_{total} = M_1 M_2$. For equal real reference impedances Z_0, define $\Delta = A + B/Z_0 + C Z_0 + D$. Then $S_{11} = (A + B/Z_0 - C Z_0 - D)/\Delta$; $S_{21} = 2/\Delta$; $S_{22} = (-A + B/Z_0 - C Z_0 + D)/\Delta$; $S_{12} = 2(AD - BC)/\Delta$.

5.5 Insertion loss and S-parameter interpretation

Insertion loss is $IL = -20 \log_{10}|S_{21}|$; return loss is $RL = -20 \log_{10}|S_{11}|$. For an ideal matched straight section with only estimated conductor attenuation, S_{21} magnitude corresponds to about 0.033 dB loss over 0.30 m, while $S_{11} = 0$ in the ideal matched model. A practical assembly will have nonzero reflection and additional transition loss.

5.6 Verification plan

Use a calibrated vector network analyser with suitable waveguide calibration standards to measure S11 and S21 over the selected band. Alternatively, model the exact guide and transition geometry in a full-wave solver, define waveguide ports and copper conductivity, and compare S-parameters with hand calculations. Report calibration plane, mesh convergence, conductor roughness, port mode, reference impedance and flange/transition geometry. Software results alone do not replace hand calculations.

6. Stage 6 – Validate, Critique and Defend the Final Design (10 marks)

6.1 Specification verification table

Requirement	Specified target	Initial design estimate	Final design status
Operating frequency / single-mode range	10 GHz; margin-based 7.22–11.80 GHz	10 GHz lies inside estimated range	Pass analytically; verify transitions
Dominant mode / cutoff	TE10 dominant; higher mode above band	fc10 ≈ 6.557 GHz; next cutoff ≈ 13.114 GHz	Pass for ideal straight WR-90
Insertion loss	≤0.5 dB over 0.30 m	Straight-section conductor estimate ≈ 0.033 dB	Full assembly not verified
Impedance / VSWR	VSWR ≤1.22 after matching	Illustrative unmatched load VSWR 1.500	Matching requires simulation/measurement
Return loss	≥20 dB after matching	Illustrative unmatched load RL 13.98 dB	Target requires S11 verification
EMI shielding / isolation	Application-specific target	Metal guide expected to shield; joints may leak	Requires EMC test
Power / size	Within component ratings; WR-90 dimensions	Standard geometry; low-power bench test recommended	Confirm ratings and dimensions with supplier

6.2 Assumptions and limitations

The analytical model assumes a straight, uniform, air-filled guide with idealized geometry and smooth copper walls. It does not fully include wall roughness, flange gaps, bends, launchers, manufacturing defects, thermal expansion, frequency-dependent material properties or radiation at discontinuities. These effects can increase loss and reflection. The equivalent load and quarter-wave example illustrate the method but do not replace a geometry-specific waveguide matching design.

6.3 Tolerance estimate

For TE10, $f_{c10} = c/(2a)$, so a small relative width change gives $\Delta f_c/f_c \approx -\Delta a/a$. A ±5% width tolerance gives approximately ∓5% cutoff variation. For a = 22.86 mm, the hypothetical width range is 21.717–24.003 mm, giving cutoff approximately 6.90–6.24 GHz. The 10 GHz operating point remains above cutoff, but the precise mode margin and matching response should be recalculated for worst-case dimensions.

6.4 EMC, safety and experimental test plan

1. Inspect guide, flanges, fasteners and transitions; check dimensional tolerances. 2. Use a low-power source and matched rated load before connecting a transmitter. 3. Calibrate a VNA with suitable waveguide standards. 4. Measure S11 and S21 over the operating band and compute return loss and insertion loss. 5. Repeat after reassembly to check flange repeatability. 6. Inspect leakage around joints using an approved method and follow institutional RF safety procedures. 7. Compare measurement with analytical estimates and document deviations.

6.5 Improved design proposal

Retain WR-90 for the stated 10 GHz application, improve flange alignment and surface condition, and design a geometry-specific matching transition. Verify the updated model with a converged EM simulation and measure a prototype. This addresses the unresolved issues of transition loss, mismatch and tolerance sensitivity.

7. Final Engineering Decision (10 marks)

Initial design: NO for unconditional adoption. The WR-90 dimensions support TE10 propagation at 10 GHz, but the complete insertion-loss target, return-loss target, transition losses and EMI performance have not been verified. The initial design is analytically plausible but does not yet satisfy the evidence requirements for a finished engineering product.

Final proposed design: CONDITIONAL YES, subject to geometry-specific matching, verified transition-loss data, VNA/full-wave validation and confirmation of the application's power and EMC requirements. The selected mode has a useful single-mode margin at 10 GHz, and standard guide dimensions are appropriate for this example. Adoption should be approved only after measured or independently simulated results show that insertion loss, return loss, VSWR, shielding and tolerance requirements are met.

Remaining non-compliance and corrective action

The main remaining gap is that complete assembly loss and S11 are not yet verified. Add a designed waveguide transition/matching feature, use measured or manufacturer-supported material/finish data, include all flanges and bends in the EM model, and run a VNA test. If the full assembly still exceeds the loss target, revise the transition geometry, reduce unnecessary length and joints, or consider another structure only after repeating the mode and loss analysis.

8. Conclusion

An analytical WR-90 rectangular waveguide design has been developed for a nominal 10 GHz X-band radar feed. Calculations identify TE10 as the dominant mode, determine modal cutoff limits, guide wavelength, phase/group velocities and TE modal impedance, and estimate copper surface resistance, skin depth and straight-guide conductor attenuation. A weighted decision matrix favours waveguide under the chosen low-loss, shielding and power-handling priorities. Network theory is applied to mismatch, VSWR, return loss, insertion loss, ABCD cascades and S-parameters. The analysis supports continued development but does not establish complete hardware compliance because transition loss and practical matching have not been measured or simulated in a geometry-specific model. Final adoption requires independent validation and a documented test report.

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[4] NIST, CODATA values of the fundamental physical constants, for c and electromagnetic constants.

[5] ECA13 – Microwave Engineering, CO1 Application-Based Assessment assignment brief, Stages 1–6 and Final Engineering Decision.

Appendix A – Calculation and simulation workflow

Recommended calculation sequence: (1) enter a, b, f, length, conductivity and target load; (2) calculate modal cutoffs; (3) verify f lies between dominant and next-mode cutoffs; (4) calculate β, λg, vp, vg, group delay and ZTE; (5) calculate skin depth, surface resistance and conductor attenuation; (6) calculate Γ, VSWR and return loss for the chosen load; (7) model the matching transition and cascade sections with ABCD matrices; (8) plot S11/S21 magnitude and phase; (9) repeat for tolerance extremes; (10) compare every result with Stage 1 acceptance criteria.

Suggested software setup: MATLAB or Python/NumPy for analytical equations and plots; a full-wave EM solver for actual transition geometry; a vector network analyser for experimental S-parameters. Record software version, material properties, boundary conditions, mesh settings, calibration method and assumptions. Do not label analytical plots as measured data.

Appendix B – Key formula sheet

Quantity	Formula
Free-space wavelength	$\lambda 0 = c/f$
Rectangular guide cutoff	$fc(mn) = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$
TE10 propagation constant	$\beta = \sqrt{[k0^2 - (\pi/a)^2]}$
Guide wavelength	$\lambda g = 2\pi/\beta$
Phase velocity	$vp = \omega/\beta$
Group velocity	$vg = c^2/vp$ for air-filled guide
TE modal impedance	$ZTE = \omega\mu/\beta = \eta 0/\sqrt{[1 - (fc/f)^2]}$
Skin depth	$\delta = \sqrt{[2/(\omega\mu\sigma)]}$
Surface resistance	$Rs = \sqrt{(\pi f\mu/\sigma)}$
Reflection coefficient	$\Gamma = (ZL - Z0)/(ZL + Z0)$
VSWR	$(1 + \lvert\Gamma\rvert)/(1 - \lvert\Gamma\rvert)$
Return loss	$RL = -20\ \log 10\lvert S11\rvert$ dB
Insertion loss	$IL = -20\ \log 10\lvert S21\rvert$ dB
Quarter-wave transformer	$Zt = \sqrt{(Z0\ ZL)}$, electrical length $\lambda/4$ at design frequency